Offline → online sync verification (#87)

Manual run of the airplane-mode sequence from #87's acceptance criteria, on the Pixel 10 Pro emulator (API 37; #128's full sign-off still needs an API 26 run). Test account: `fundtrail.offline.test@example.com` (created for this run only).

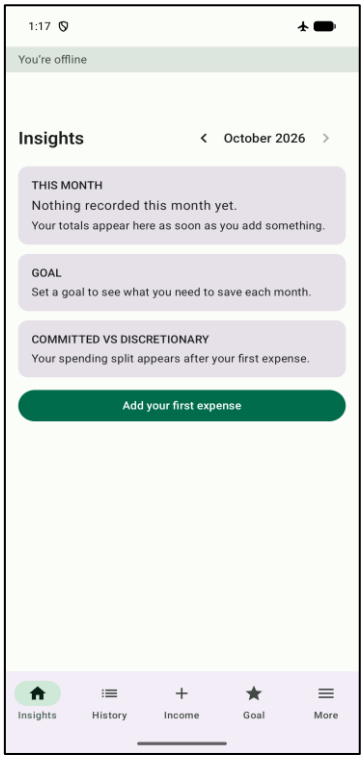
Steps and evidence

1. **Register a new user.** Sign-up creates the Firebase Auth account and the 4 default accounts / default categories. (screenshot 1)
2. **Airplane mode on.** The "You're offline" banner appears immediately. (screenshot 2)
3. **Add an expense while offline** (Rent, LKR 1,112.00, Main bank). Saved and visible in the Insights dashboard well under 1 second after tapping Save (screenshot 3), and immediately in History too (screenshot 4).
4. **Force-stop the app**, then reopen it **while still offline**. The entry is still there — it was written to Room, not just held in memory. (screenshot 5)
5. **Reconnect.** The offline banner clears. (screenshot 6)
6. **Verify the write reached Firestore.** Rather than extract any credential or token to query Firestore directly, this was verified through the app's own flow: sign out (clears Room per the M1 logout rule — architecture.md, #51) while online, then sign back in with the same account. Room starts empty again, so anything that reappears had to come from Firestore's `users/{uid}/transactions` collection. The Rent entry reappears after the fresh sign-in, confirming the pending write synced while offline and was pulled back down cleanly. (screenshot 7)

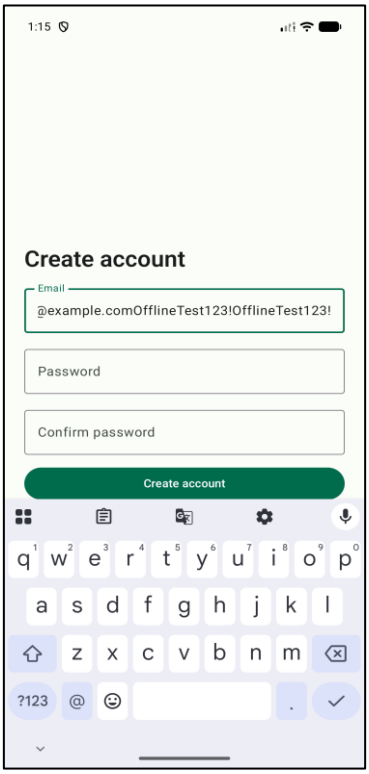
Screenshots

Screenshots (images not embedded)

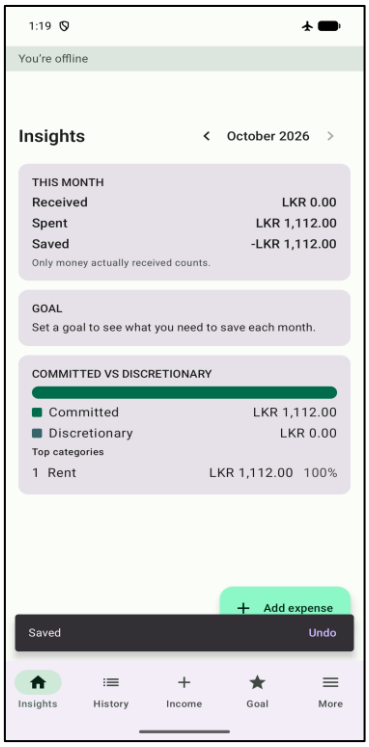
1. Create account



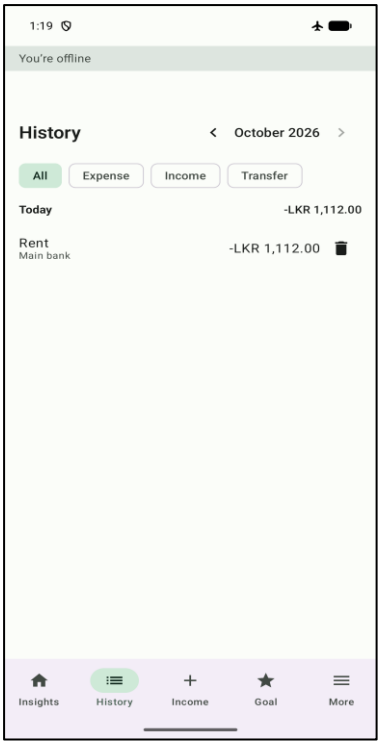
2. Offline banner



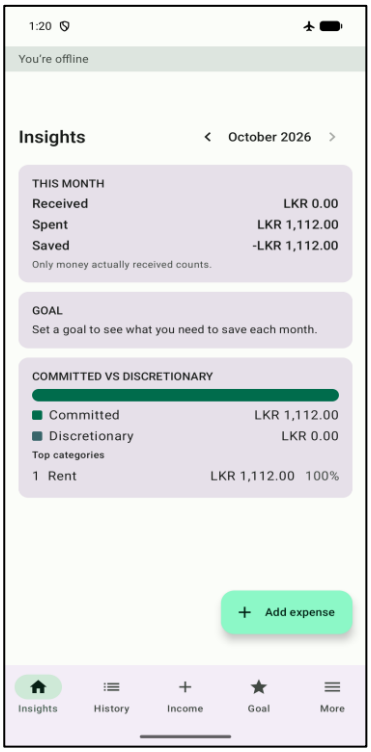
3. Saved under 1s



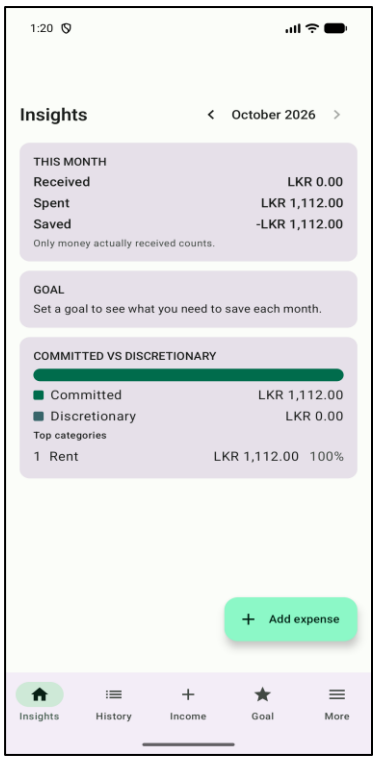
4. History offline



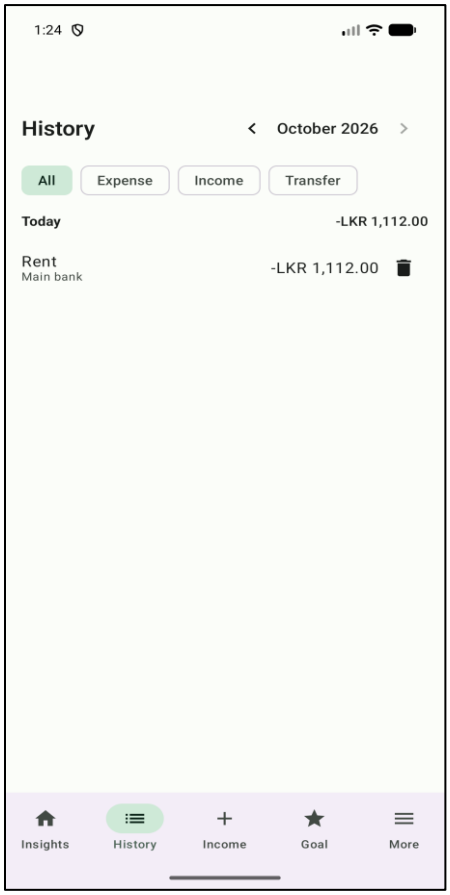
5. Reopen offline



6. Reconnected



7. Synced from Firestore after fresh sign-in



Result

All three AC bullets hold:

- `DocumentChange` / Room write-through: the expense is visible in the dashboard and History within 1s of Save, while offline.
- Force-stop + reopen offline: the entry survives, proving it's in Room, not just view state.
- Reconnect: the entry is present in Firestore (`users/{uid}/transactions`), verified via sign-out/sign-in round-trip rather than a direct console/API query.

Test account `fundtrail.offline.test@example.com` and its data are safe to delete; they exist only in the dev Firebase project for this test run.